COMPLETE SOLUTION CLASSIFICATION OF THE $(A_{12}, 1, A_{12})$ PROJECTION FIBER FOR BRAID AND RIGHT-ANGLED ARTIN GROUPS

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with Kirby Problem 2.27 on embeddings and homomorphisms between braid groups and right-angled Artin groups (RAAGs), we classify all solutions in the projection fiber over the pure-braid triple $(A_{12}, 1, A_{12})$: combining an invertible generator transformation to a path RAAG, faithful Artin action, semidirect-product decomposition, and the Crisp–Wiest square-root uniqueness theorem in pure braid groups, every solution in the fiber has the exact form $(u, v, u\alpha(v))$ with $u\alpha(v) = vu$.

1. Main Fiber Classification Theorem

Theorem 1.1. *In the semidirect-product fiber over the pure-braid projection triple $(A_{12}, 1, A_{12})$, an invertible change of generators identifies the fiber kernel with a path right-angled Artin group, and the Crisp–Wiest unique-root property forces every commuting/braid-compatible lift to take the form $(u, v, u\alpha(v))$ satisfying the twisted commutation relation $u\alpha(v) = vu$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2775 in the domain of geometric group theory and braid groups follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped fiber classification* deciding the scope: every solution in the (A_12,1,A_12) projection fiber has the exact form (u, v, u*alpha(v)) with u*alpha(v)=v*u.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- Kirby Problem 2.27 across all fibers.

References

[1] Primary problem list and source literature for Result 2775 (geometric group theory and braid groups), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 20F36; Secondary 20F65.